

Hypothesis test for $P_1 - P_2$.

$$H_0: P_1 - P_2 = 0 \quad \text{vs.} \quad \begin{cases} H_a: P_1 - P_2 \neq 0 & \text{two-sided} \\ H_a: P_1 - P_2 > 0 & \text{right-sided} \\ H_a: P_1 - P_2 < 0 & \text{left-sided} \end{cases}$$

1) Test statistic.

$$\hat{p} = \frac{x_1 + x_2}{n_1 + n_2} \quad (\text{overall proportion since } H_0: P_1 = P_2 = P.)$$

$$Z^* = \frac{\hat{P}_1 - \hat{P}_2}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

2) critical value.

$$\text{two-sided} : Z_{\frac{\alpha}{2}}$$

$$\text{right-sided} : Z_{\alpha}$$

$$\text{left-sided} : -Z_{\alpha}$$

3) Decision

$$\text{two-sided} : \text{If } |Z^*| > Z_{\frac{\alpha}{2}}, \text{ then reject } H_0$$

$$\text{right-sided} : \text{If } Z^* > Z_{\alpha}, \text{ then reject } H_0$$

$$\text{left-sided} : \text{If } Z^* < -Z_{\alpha}, \text{ then reject } H_0$$

4) P-value.

$$\text{two-sided} : p\text{-value} = 2P(Z > |Z^*|)$$

$$\text{right-sided} : p\text{-value} = P(Z > Z^*)$$

$$\text{left-sided} : p\text{-value} = P(Z < Z^*)$$

5) CI for $P_1 - P_2$

$$(\hat{P}_1 - \hat{P}_2) \pm Z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{P}_1(1-\hat{P}_1)}{n_1} + \frac{\hat{P}_2(1-\hat{P}_2)}{n_2}}$$